

Presentation Script

Read-aloud text for every slide of *DSDR_Final_Presentation.pptx*

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How to use this. Each block is one slide: the slide number, a small picture of the slide, and the paragraphs to read, numbered **Para 1, Para 2, ...** Read them in order, at a normal pace, and change the slide when the block ends. The brown lines ( *Point to*) are not read: they say where to point on the screen before the next paragraph.

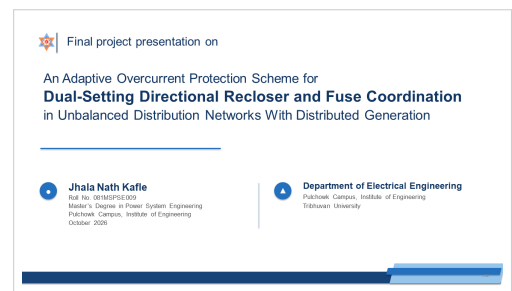
Timing. Read everything: about **22 minutes**. Skip every paragraph marked *optional* (printed in grey): about **19 minutes**. Both are at a calm reading speed, for slides 1 to 27 and the closing slide. The talk still makes sense without the optional paragraphs.

Numbers. Read “0.121” as “zero point one two one”, “4.16 kV” as “four point one six kilovolts”, “400E” as “four hundred E”, and “CTI” as the three letters.

SLIDE 1

Title

about 30 s to read



Para 1

Good afternoon, respected Head of the Department, respected coordinator, teachers and friends. My name is Jhala Nath Kafle, roll number 081 MSPSE 009.

Para 2

My project is an adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in unbalanced distribution networks with distributed generation. In short: how to keep the recloser and the fuses of a feeder coordinated after a generator is connected to it.

SLIDE 2

Presentation Outline

about 25 s to read



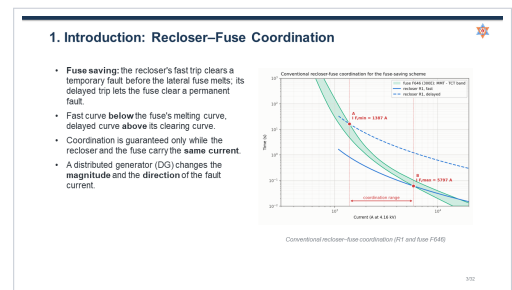
Para 1 This is the outline. I will start with the introduction and the problem, then the objectives and the scope. After that I explain the methodology, then the results, and I close with the conclusions and the references.

Para 2 The comparison with the reference paper is kept in the appendix, after the last slide.
optional

SLIDE 3

1. Introduction: Recloser-Fuse Coordination

about 1 min 00 s to read



Para 1 Most faults on overhead distribution lines are temporary. In a fuse-saving scheme the recloser trips first, on its fast curve, so a temporary fault is cleared and the fuse is not lost. If the fault is permanent, the recloser moves to its delayed curve, and the fuse then clears only its own lateral.

▷ *Point to: the graph on the right: solid blue line = fast curve, green band = fuse, dashed blue = delayed curve*

Para 2 On the graph, the fast curve of the recloser lies below the fuse band, and the delayed curve lies above it. Coordination exists only in the range of current between the two marked points.

Para 3 This works only while the recloser and the fuse carry the same current. A distributed generator changes both the magnitude and the direction of the fault current, and that is where the problem starts.

SLIDE 4

1. Introduction: Problem Statement

about 1 min 15 s to read

1. Introduction: Problem Statement

Without DG
Fault current flows from the substation to the fault; recloser and fuse see the same current.

With DG
The fuse carries grid + DG current; the recloser only the grid share. A mid-line recloser sees reverse current.

Consequence
The fuse can melt before the fast trip; fuse saving is lost.

- **Problem:** loss of recloser–fuse coordination when a DG is connected; a single fixed setting does not suit both the forward (grid) and the reverse (DG) fault current.
- **Proposed solution:** a **Dual-Setting Directional Recloser (DSDR)** – the mid-line recloser R2 gets independent forward and reverse settings, selected by the direction of the current.
- **This project:** implements and tests the DSDR method on the IEEE 13-node feeder in DIgSILENT PowerFactory with the manufacturer-based time–current characteristics of commercial relays and fuses.

▷ *Point to: the three boxes, from left to right*

Para 1

Without DG, the fault current flows from the substation to the fault, so the recloser and the fuse see the same current. With a DG, the fuse carries the grid current plus the DG current, but the recloser carries only the grid share. A recloser in the middle of the line can even see current in the reverse direction.

Para 2

The consequence is that the fuse can melt before the fast trip, and fuse saving is lost.

Para 3

A single fixed setting cannot suit both the forward current from the grid and the reverse current from the DG. The solution studied here is the dual-setting directional recloser, in short DSDR. The mid-line recloser, R2, gets two independent groups of settings, one forward and one reverse, and the direction of the current selects the group.

Para 4

In this project I implemented and tested this method on the IEEE 13-node feeder in DIgSILENT PowerFactory, with the manufacturers' characteristics of commercial relays and fuses.

SLIDE 5

2. Objectives

about 1 min 00 s to read

2. Objectives

General Objective

To build and validate a DigSILENT PowerFactory implementation of the dual-setting directional recloser (DSDR) coordination method on the IEEE 13-node feeder, with and without distributed generation.

Specific Objectives

1. Build and verify the IEEE 13-node feeder with its recloser and fuse protection.
2. Perform unbalanced load-flow and short-circuit studies for LG, LL, LLL and LLL faults.
3. Calculate the recloser pickups, time dials and fuse coefficients.
4. Establish the coordination without DG, then with a 4.05 MVA synchronous DG at node 692.
5. Design the DSDR settings of R2, classify the resulting coordination, find the operating times.
6. Verify the reclosing sequence in the time domain (EMT simulation).
7. Study the effect of DG penetration from 0 to 100 %.

- Para 1** The general objective is to build and validate a PowerFactory implementation of the DSDR coordination method on the IEEE 13-node feeder, with and without distributed generation.
- Para 2** There are seven specific objectives. First, to build and verify the feeder model with its recloser and fuse protection. Second, to carry out unbalanced load-flow and short-circuit studies for all fault types. Third, to calculate the recloser pickups, the time dials and the fuse coefficients.
- Para 3** Fourth, to establish the coordination without DG, and then with a 4.05 MVA synchronous DG at node 692. Fifth, to design the DSDR settings of R2 and classify the resulting coordination. Sixth, to verify the reclosing sequence in the time domain. And seventh, to study the effect of the DG penetration level.

SLIDE 6

3. Scope and Limitations

about 1 min 10 s to read

3. Scope and Limitations

Scope

- IEEE 13-node feeder, 4.16 kV, unbalanced.
- LG, LL, LLL and LLL faults at 12 locations: 39 node/fault-type cells.
- Pickup, time-dial and fuse coordination calculations.
- DSDR forward and reverse settings; coordination with and without the DSDR.
- One EMT simulation of the reclosing sequence; DG penetration 0–100 %.

Limitations

- The method does not uniquely specify the R2 settings, fuse sizes or fault type of each operating-time case; they were defined in this study and documented.
- The CDG34 library relay is not directional: two relay units, direction from the fault location.
- Zero-margin criterion: $CTI = I_{MMT} - I_F > 0$; practical margins evaluated separately.
- One DG location (692); IEEE 34-node feeder not modelled.

▷ *Point to: the left column, Scope*

- Para 1** The study covers the IEEE 13-node feeder at 4.16 kV, which is an unbalanced feeder. Four fault types are applied at twelve locations. Together this gives 39 combinations of node and fault type, which I call cells. The study also includes one EMT simulation of the reclosing sequence and a study of DG penetration from zero to 100 per cent.

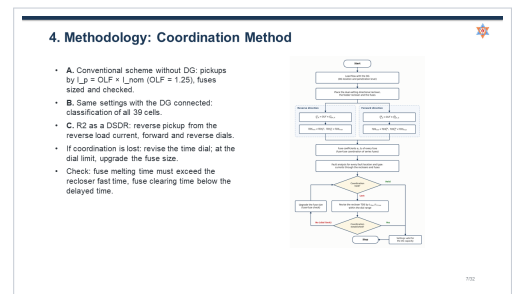
▷ *Point to: the right column, Limitations*

- Para 2** There are four limitations. First, the method does not uniquely specify the settings of R2 or the fuse sizes, so these were defined in this study and documented. Second, the CDG34 relay model in the library is not directional, so the two setting groups are modelled as two relay units.
- Para 3** optional Third, the coordination is classified with a zero margin, and practical margins are evaluated separately. Fourth, only one DG location is studied, and the IEEE 34-node feeder is not modelled.

SLIDE 7

4. Methodology: Coordination Method

about 1 min 00 s to read



▷ *Point to: the flowchart on the right, top to bottom*

Para 1

The method has three stages. Stage A is the conventional scheme without DG. The pickup of each recloser is 1.25 times the rated current of its branch, and the fuses are sized and checked.

Para 2

Stage B keeps the same settings and connects the DG. All 39 cells are classified again. Stage C makes R2 a DSDR: the reverse pickup is taken from the reverse load current, and forward and reverse dials are set.

Para 3

If coordination is lost, the time dial is revised first. If the dial is already at its limit, the fuse size is upgraded. The check is always the same: the fuse must not start to melt before the fast trip, and it must clear before the delayed trip.

SLIDE 8

4. Methodology: Mathematical Formulation

about 1 min 10 s to read

4. Methodology: Mathematical Formulation of the Method

1 Recloser operating time (forward fv, reverse rv)

$$t_{f,rv}^{cl} = TDS_{cl} \left[\frac{A}{(I_{f,rv}^{cl})^2 - 1} + B \right]$$

$$t_{f,D}^{cl} = TDS_{cl,D} \left[\frac{A}{(I_{f,D}^{cl})^2 - 1} + B \right]$$

$d \in \{fv, rv\}$: F = fast, D = delayed; one curve, two time dials

2 Pickup and constraints

$$I_{f,D}^{cl} = OLF \times I_{f,D,max}$$

$$I_{f,D}^{cl} = OLF \times I_{f,D,max}$$

$$TDS_{f,D} = TDS_{f,D}^{cl}, TDS_{f,D}^{cl} < TDS_{f,D,max}$$

$$I_{f,D,max} < I_{f,D} < I_{f,D,max} \quad (OLF = 1.25)$$

3 Fuse characteristic and coefficient

$$\log t_{fuse,i} = a \log I_{f,i} + b_i \quad (a_i = -1.8)$$

$$b_i = \log \left[\frac{t_{f,i} + t_{D,i}}{2} \right] - a \log I_{f,i} \quad (\text{single fuse})$$

$$b_i = \log \left[t_{f,i} + \frac{(t_{D,i} - t_{f,i})}{2} \right] - a \log I_{f,i} \quad (z \text{ in series})$$

4 Coordination conditions (fuse saving)

$$CTI = t_{fuse}(I_{f,max}) - t_{f,rv}(I_{f,max}) > 0 \quad \text{fast trip first}$$

$$t_{f,D}(I_{f,max}) < t_{f,D,max} \quad \text{fuse clears first}$$

$$t_{f,D,i} \leq 0.75 t_{fuse,i+1} \quad \text{series fuses}$$

▷ Point to: box 1, top left

Para 1

These are the equations of the method. Box one is the operating time of the recloser: one inverse-time curve, used with two time dials, one for the fast and one for the delayed operation, and written separately for the forward and the reverse direction.

▷ Point to: box 2, top right

Para 2

optional

Box two gives the pickup, which is the overload factor times the rated current, and the limits of the dial and of the pickup.

▷ Point to: box 3, bottom left

Para 3

Box three is the fuse. On logarithmic axes the fuse curve is a straight line with slope a equal to minus 1.8. The coefficient b places the fuse between the fast and the delayed time of the recloser.

▷ Point to: box 4, bottom right

Para 4

Box four gives the conditions for fuse saving. The coordination time interval, CTI, is the melting time of the fuse minus the fast time of the recloser, and it must be positive. The fuse must also clear before the delayed trip.

SLIDE 9

4. Methodology: Operating Times from the Manufacturers' Curves

about 1 min 05 s to read

4. Methodology: Operating Times from the Manufacturers' Curves

Method: $t = TDS \left[\frac{A}{(M-C)^n} + B \right]$ → in general: $t = TDS \times g(M)$, $M = I_f/I_p$, $g(M) = \text{curve of the specific relay}$

R1: GE IAC78801A, extremely inverse

$$g(M) = A + \frac{B}{M - C} + \frac{D}{(M - C)^2} + \frac{E}{(M - C)^3}$$

$A = 0.004$, $B = 0.6379$, $C = 0.62$, $D = 1.7872$, $E = 0.2461$

GE Multilin manual: $I_p = 720$ A, $TDS = 0.5 / 10$

R2 (DSDR): GE/Alstom CDG34, extremely inverse

$$g(M) = C \left(\frac{I_p}{I_f} \right)^n \cdot f = \frac{\ln(M/M_0)}{\ln(M_0/M_1)}$$

manufacturer table (PowerFactory library), $M = I_f/I_p$,
feed I_p 300/600 A, rev 150/300 A, TMS 0.1 / 1.0

Why the manufacturers' curves

- The method fits the structure $t = TDS \times g(M)$. A, B and n only describe a generic curve.
- R1 (GE IAC, dial called TDS) and R2 (CDG34, dial called TMS) use the curves of their manufacturers, both scale the curve in the same way.

How we know it is right

- R1 reproduces the paper's worked example: $0.096 / 1.926$ s against $0.097 / 1.932$ s (4219 A).
- PowerFactory's own relay models confirm all 1896 operating times within 1.22 %.

Para 1 The method writes the operating time as the time dial multiplied by a function of M, where M is the fault current divided by the pickup. The constants A, B and n describe only a generic curve.

▷ Point to: the two blue boxes: R1 on the left, R2 on the right

Para 2 So I kept this structure, and used the manufacturer's curve of each relay. For R1 it is the GE IAC extremely inverse equation. For R2 it is the manufacturer's table of the CDG34 relay. The dial is called TDS for the IAC relay and TMS for the CDG34; both scale the curve in the same way.

▷ Point to: bottom right, How we know it is right

Para 3 This was checked in two ways. R1 reproduces the worked example of the reference optional paper: 0.096 and 1.926 seconds, against 0.097 and 1.932 seconds. And PowerFactory's own relay models confirm all 1896 operating times within 1.22 per cent.

SLIDE 10

4. Methodology: Test System in PowerFactory

about 45 s to read

4. Methodology: Test System in PowerFactory

- IEEE 13-node feeder, 4.16 kV, 115 kV grid via 5 MVA transformer.
- DG: 4.05 MVA synchronous machine at node 692, 0.694 16 kV transformer.
- R1: GE IAC78801A at the feeder head (R060-632).
- R2: CDG34 on line 632-671 (the DSDR).
- Fuses: A055C (E-rated) on the laterals.
- Unbalanced load flow, short-circuit and EMT studies in DigSILENT PowerFactory 2021.

▷ Point to: the single-line diagram on the left

Para 1 This is the model in PowerFactory. It is the IEEE 13-node feeder at 4.16 kV, supplied from a 115 kV grid through a 5 MVA transformer.

Para 2 The DG is a 4.05 MVA synchronous machine at node 692. R1 is a GE IAC relay at the head of the feeder. R2 is a CDG34 relay on the line from 632 to 671, and this is the recloser that becomes the DSDR. The laterals are protected by A055C fuses.

Para 3 Unbalanced load flow, short-circuit and EMT studies were all carried out on this one optional model.

SLIDE 11

4. Methodology: Parameters of the Synchronous DG

about 30 s to read

4. Methodology: Parameters of the Synchronous DG					
Parameter	Symbol	Value	Parameter	Symbol	Value
Leakage reactance	X_l	0.05 pu	q-axis subtransient reactance	X''_q	0.115 pu
Stator resistance	R_a	0.0014 pu	q-axis transient OC time const.	T'_{dq}	1.25 s
d-axis synchronous reactance	X_d	1.4 pu	q-axis subtransient OC time const.	T''_{dq}	0.19 s
d-axis transient reactance	X'_d	0.231 pu	Mechanical starting time	$M = 2H$	1.5 s
d-axis subtransient reactance	X''_d	0.118 pu	DG rating	S_n	4.05 MVA
d-axis transient OC time const.	T'_{d0}	5.5 s	DG voltage	U_n	0.69 kV
d-axis subtransient OC time const.	T''_{d0}	0.05 s	Step-up transformer		0.69 / 4.16 kV
q-axis synchronous reactance	X_q	1.372 pu	Transformer leakage reactance	x_l	0.15 pu
q-axis transient reactance	X'_q	0.8 pu	Connection node		692

Dynamic synchronous machine in PowerFactory, operating point 2.24 RM, voltage control at 1.0 pu

Para 1 These are the parameters of the synchronous DG. It is rated 4.05 MVA at 0.69 kV and is connected at node 692 through a step-up transformer of 0.69 to 4.16 kV.

Para 2 optional It is modelled as a dynamic synchronous machine with its subtransient, transient and synchronous reactances, so its fault contribution is represented correctly. It operates at 3.24 megawatts.

SLIDE 12

4. Methodology: Protection Settings

about 55 s to read

4. Methodology: Protection Settings			
Device	Characteristic	Pickup / plug (fast / delayed)	Dial (fast / delayed)
R1 (feeder head)	GE IAC extremely inverse	720 A	TDS 0.5 / 10
R2 forward (grid current)	CDG34	300 / 600 A	TMS 0.1 / 1.0
R2 reverse (DG current)	CDG34, CT 500/5	150 / 300 A	TMS 0.1 / 1.0

- Reverse load current of R2 with the DG: 252 A → reverse pickup 315 A (OLF = 1.25).
- Dial revision could not help: R1 and R2 are already at the ends of their dial ranges.
- Fuse revision (step 9 of the method): F632 → 500E; F633, F646, F-DL, F671-1 → 400E; F662-R → 250E.

▷ Point to: the table, row by row

Para 1 These are the settings. R1 at the feeder head has a pickup of 720 amperes and time dials of 0.5 for the fast and 10 for the delayed operation. R2 forward uses plug settings of 300 and 600 amperes. The new reverse group of R2 uses 150 and 300 amperes.

Para 2 optional With the DG connected, the load current through R2 reverses and becomes 252 amperes. With the overload factor this gives a reverse pickup of 315 amperes.

Para 3 A revision of the dials could not help, because R1 and R2 are already at the ends of their dial ranges. So the fuse sizes were revised, which is step nine of the method, as listed in the last line.

SLIDE 13

5. Results: Branch Currents and Fault Levels (DG out)

about 35 s to read

5. Results: Branch Currents and Fault Levels (DG out)			
Branch	Rated current I _{nom} (A)	I _{min} (kA) LG 3 Ω, farthest node	I _{max} (kA) bolted, nearest node
B000-032	587.7	1.07	4.73
B32-033	81.0	0.78	4.1
B32-045	143.1	0.73	3.42
B32-071	470.0	0.98	3.39
XFMR 1 HV side	81.0	0.08	1.81
XFMR 1 LV side	708.4	0.71	15.67
B45-046	66.1	0.73	3.11
B45-080	239.7	0.74	3.36
B71-084	71.1	0.95	2.95
B71-080	0.0	0.64	2.93
B82-075	205.9	0.74	3.15
B84-011	71.1	0.88	1.86
B84-052	63.0	0.96	1.87

Unbalanced load flow (largest phase) and short circuit, complete method, XFMR 1 LV side at 0.48 kV

Para 1 Now the results. The first is the base case without DG.

▷ Point to: the columns, from left to right

Para 2 For every protected branch the table gives the rated current from the load flow, the minimum fault current, and the maximum fault current. The minimum is a line-to-ground fault through 3 ohms at the farthest node. The maximum is a bolted fault at the nearest node.

Para 3 The feeder head carries 588 amperes of load, and its maximum fault current is 4.73 optional kiloamperes.

SLIDE 14

5. Results: Fuse Coefficients

about 30 s to read

5. Results: Fuse Coefficients (eq. 9, a = -1.8)								
Fuse	I _{fast} (A)	I _{slow} (A)	t _{fast} (s)	t _{slow} (s)	b	Installed fuse	t _{fast} (s)	t _{slow} (s)
F032	349	25	0.13	2.07	1.024	60E	0.02	0.512
F033	4102	151	0.100	2.000	1.054	250E	0.02	0.162
F034	15071	102	0.439	8.772	3.217	800E	0.06	0.021
F045	349	102	0.13	2.07	0.979	250E	0.35	0.158
F046	3112	102	0.106	2.110	1.142	300E	0.273	0.223
F045-R	3940	151	0.106	2.130	1.118	250E	0.112	0.112
F071	3380	151	0.054	1.001	0.827	250E	0.27	0.161
F071-1	3380	351	0.054	1.001	1.254	300E	0.278	0.278
F080	3380	102	0.054	1.001	0.509	250E	0.11	0.084
F080-R	3151	201	0.057	1.029	0.943	250E	0.27	0.112
F075	3151	151	0.057	1.029	0.930	250E	0.112	0.112
F071-2	3055	351	0.060	2.122	1.346	300E	0.428	0.428
F084	1807	201	0.113	5.778	2.946	250E	0.465	0.465
F011	1807	151	0.113	5.778	1.530	150E	0.120	0.120
F052	1874	102	0.112	5.680	1.381	150E	0.107	0.107

: largest fault current below the fuse, DG out. I_{fast} / I_{slow} position in a series of 2 fuses (I_{fast} = obtained to the fast)

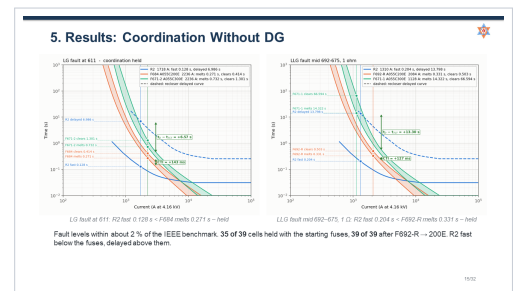
Para 1 This table gives the fuse coefficients from the equation of the method, with the slope a equal to minus 1.8.

Para 2 For each fuse, the coefficient b follows from the fast and the delayed time of the recloser at the largest fault current below that fuse. The last two columns give the fuse that is actually installed and its melting time at the same current.

SLIDE 15

5. Results: Coordination Without DG

about 55 s to read



Para 1

Without DG, the fault levels of the model agree with the IEEE benchmark within about 2 per cent. 35 of the 39 cells are coordinated with the starting fuse sizes, and all 39 after one fuse is changed.

▷ Point to: the left plot, then the right plot

Para 2

Two examples. On the left, a line-to-ground fault at node 611: R2 trips in 0.128 seconds, and the fuse would start to melt only at 0.271 seconds. On the right, a double line-to-ground fault between 692 and 675: R2 trips in 0.204 seconds, before the fuse melts at 0.331 seconds.

Para 3

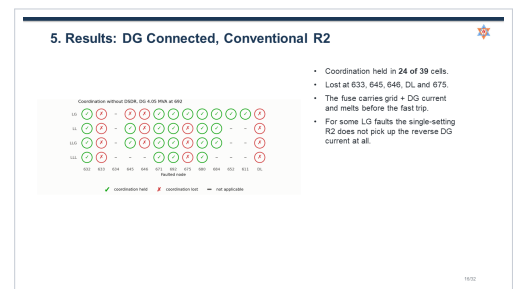
optional

In both, the fast curve of R2 is below the fuses and its delayed curve is above them. So without DG the conventional scheme works.

SLIDE 16

5. Results: DG Connected, Conventional R2

about 35 s to read



▷ Point to: the chart: green tick = coordination held, red cross = lost

Para 1

Now the DG is connected and the settings are not changed. Coordination holds in only 24 of the 39 cells.

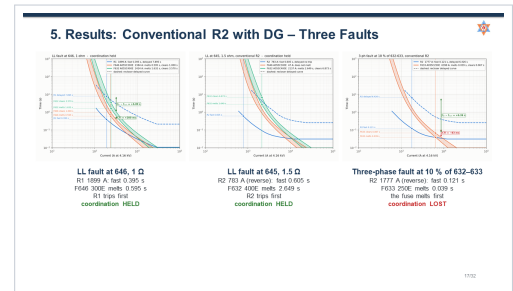
Para 2

It is lost at nodes 633, 645, 646, the distributed load and 675. The reason is the one from the introduction: the fuse carries the grid and the DG current, and it melts before the fast trip. For some line-to-ground faults the single-setting R2 does not even pick up the reverse current from the DG.

SLIDE 17

5. Results: Conventional R2 with DG - Three Faults

about 45 s to read



Para 1 These are three faults with the DG and the conventional settings.

▷ Point to: the left plot, the middle plot, then the right plot

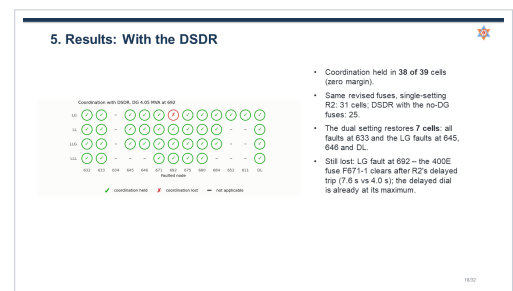
Para 2 On the left, a line-to-line fault at 646 through 1 ohm: R1 trips first, and coordination is held. In the middle, a line-to-line fault at 645: R2 carries 783 amperes in reverse and still trips first, so it is also held.

Para 3 On the right, a three-phase fault close to node 632. R2 carries 1777 amperes in reverse and trips in 0.121 seconds. But the fuse F633 melts already at 0.039 seconds. The fuse melts first, and coordination is lost.

SLIDE 18

5. Results: With the DSDR

about 55 s to read



▷ Point to: the chart: only one red cross is left

Para 1 Now R2 is a DSDR and the fuses are revised. Coordination is held in 38 of the 39 cells, under the zero-margin criterion.

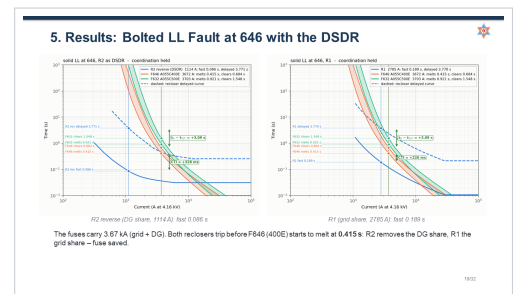
Para 2 With the same revised fuses but a single-setting R2, only 31 cells are held. So the dual setting itself restores seven cells: all the faults at 633, and the line-to-ground faults at 645, 646 and the distributed load. The DSDR with the old fuses gives only 25, so it needs the fuse revision.

Para 3 One cell stays lost: the line-to-ground fault at 692. There the 400E fuse clears in 7.6 seconds, after R2's delayed trip at 4.0 seconds, and the delayed dial of R2 is already at its maximum.

SLIDE 19

5. Results: Bolted LL Fault at 646 with the DSDR

about 45 s to read



Para 1

This is one fault in detail: a bolted line-to-line fault at 646, with the DSDR.

▷ Point to: the left plot for R2, the right plot for R1

Para 2

The fuses carry 3.67 kiloamperes, which is the grid plus the DG current. R2 sees only the DG share, 1114 amperes in the reverse direction, and its reverse group trips in 0.086 seconds. R1 sees the grid share, 2785 amperes, and trips in 0.189 seconds.

Para 3

Both reclosers trip before the fuse F646 starts to melt at 0.415 seconds. R2 removes the DG share and R1 removes the grid share, so the fuse is saved.

SLIDE 20

5. Results: Operating Times with the DSDR

about 35 s to read

Node	Fault	I R1 (A)	R1 fast / delayed (s)	I R2 (A)	R2 fast / delayed (s)	Fuse	I fuse (A)	Fuse MNT / TCT (s)
632	LL	4734	0.001 / 1.627	1013 rev	0.001 / 1.416	—	—	—
633	LL	3870	0.100 / 2.111	1000 rev	0.000 / 2.001	F633	5426	0.149 / 0.236
645	LL	3124	0.100 / 2.095	1202 rev	0.075 / 1.002	F632	4191	0.613 / 0.870
646	LL	2785	0.100 / 2.170	1114 rev	0.060 / 2.171	F646	3672	0.415 / 0.684
725	LL	3854	0.100 / 2.123	1015 rev	0.040 / 1.573	F624	5602	0.121 / 0.193
671	LL	3412	0.134 / 2.677	3381 fwd	0.054 / 1.000	—	—	—
682	LL	3412	0.134 / 2.677	3381 fwd	0.054 / 1.000	F671-1	3381	0.525 / 0.881
675	LL	3080	0.159 / 3.112	3050 fwd	0.009 / 1.941	F682-1	5022	0.005 / 0.106
684	LL	2543	0.222 / 4.437	2443 fwd	0.072 / 2.813	F671-2	3903	0.148 / 0.230
690	LL	2709	0.100 / 2.191	2707 fwd	0.000 / 2.301	—	—	—
611	LG	1814	0.420 / 0.709	1718 fwd	0.120 / 0.986	F684	2236	0.271 / 0.414
682	LG	1786	0.446 / 0.978	1761 fwd	0.123 / 0.576	F682	2239	0.136 / 0.193

Bolted LL, LL at three-phase, LG at one-phase nodes. Every fuse starts to melt after the last fast trip: smallest margins 6 ms (675) and 15 ms (64).

Para 1

This table gives the operating times with the DSDR and the DG in service, for a bolted fault at every node.

▷ Point to: the R2 current column: rev for the upper rows, fwd for the lower rows

Para 2

optional

R2 operates in the reverse direction for the nodes upstream of it, and in the forward direction for the nodes downstream.

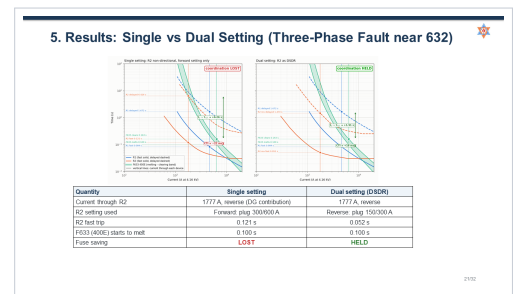
Para 3

In every row with a fuse, the fuse starts to melt after the last fast trip. The smallest margins are 6 milliseconds at node 675 and 15 milliseconds at the distributed load.

SLIDE 21

5. Results: Single vs Dual Setting

about 40 s to read



Para 1 This slide compares the single setting and the dual setting on the same fault: the three-phase fault near node 632, with the revised fuses.

▷ *Point to: the table under the plots: middle column, then right column*

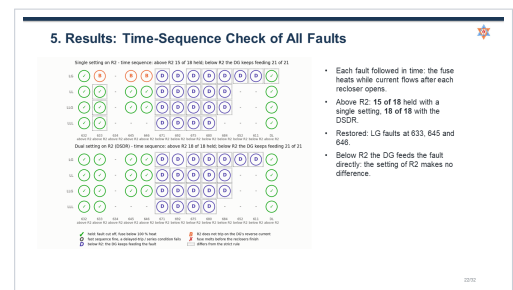
Para 2 In both cases R2 carries 1777 amperes in reverse. With the single setting, R2 uses its forward plug and trips in 0.121 seconds, but the fuse starts to melt at 0.100 seconds: fuse saving is lost.

Para 3 With the dual setting, R2 uses its reverse group and trips in 0.052 seconds, before the fuse: fuse saving is held. Only the setting group has changed.

SLIDE 22

5. Results: Time-Sequence Check of All Faults

about 40 s to read



Para 1 As a second check, every fault was followed in time. The fuse heats while current flows, also after one recloser has opened.

▷ *Point to: the upper chart = single setting, the lower chart = DSDR*

Para 2 For the 18 cells upstream of R2, the single setting holds 15, and the DSDR holds all 18. The three restored cells are the line-to-ground faults at 633, 645 and 646.

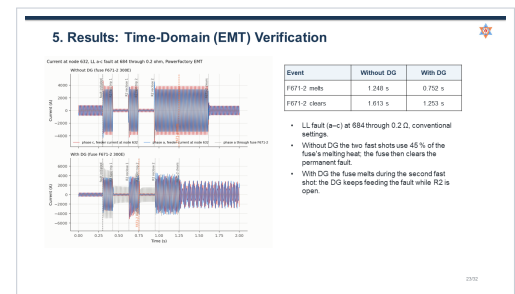
Para 3 Downstream of R2 the DG feeds the fault directly, so the setting of R2 makes no difference there. This is a limit of the method.

optional

SLIDE 23

5. Results: Time-Domain (EMT) Verification

about 50 s to read



Para 1 This is the time-domain verification with an EMT simulation. The fault is line-to-line at node 684 through 0.2 ohm, with the conventional settings.

▷ *Point to: the upper waveform = without DG, the lower waveform = with DG*

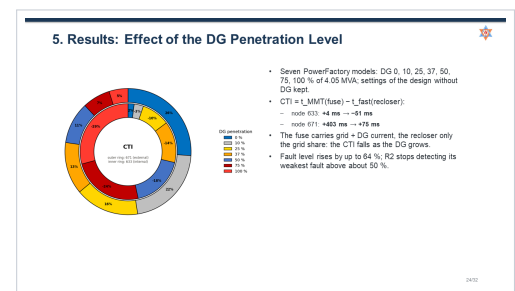
Para 2 Without DG, R2 makes two fast shots. They use only 45 per cent of the fuse's melting heat, so the fuse survives. As the fault is permanent, the fuse then melts at 1.248 seconds and clears its lateral. This is the intended sequence.

Para 3 With the DG, the fuse melts at 0.752 seconds, during the second fast shot. The fuse carries more current, and the DG keeps feeding the fault while R2 is open. Fuse saving is lost.

SLIDE 24

5. Results: Effect of the DG Penetration Level

about 1 min 00 s to read



Para 1 The last result is the effect of the DG penetration level. Seven models were made, with the DG at 0, 10, 25, 37, 50, 75 and 100 per cent of 4.05 MVA. The settings of the design without DG were kept.

▷ *Point to: the ring chart: outer ring = node 671, inner ring = node 633*

Para 2 The CTI is the melting time of the fuse minus the fast time of the recloser. At node 633 it falls from plus 4 milliseconds to minus 51 milliseconds. At node 671 it falls from plus 403 to plus 75 milliseconds.

Para 3 The reason is that the fuse carries the grid and the DG current, while the recloser carries only the grid share. The fault level rises by up to 64 per cent, and above about 50 per cent penetration R2 stops detecting its weakest fault.

SLIDE 25

5. Results: Summary and Verification

about 50 s to read

5. Results: Summary and Verification	
Stage	Cells held (of 39)
No DG, starting fuse sizes	35
No DG, after the fuse revision	39
DG connected, conventional R2	24
DG connected, no-DG fuses, R2 as DSDR	25
DG connected, revised fuses, conventional R2	31
DG connected, revised fuses, R2 as DSDR	38

- PowerFactory's own relay and fuse models reproduce the calculated operating times within 1.22 % (1896 operating times)
- These counts use the zero-margin criterion, $CTI = I_{MMT} - I_F > 0$.
- With a three-cycle breaker interrupting time, 31 cells held, with a fuse safety margin as well, 26.

▷ Point to: the table, from the top row to the bottom row

Para 1

This table summarises the results. Without DG, 35 cells are held with the starting fuses and 39 after the fuse revision. With the DG and a conventional R2, 24. With the DSDR alone, 25. With the fuse revision alone, 31. And with the DSDR and the fuse revision together, 38 of 39.

Para 2

These counts use the zero-margin criterion. With a three-cycle breaker interrupting time, 31 cells are held, and with a fuse safety margin as well, 26.

Para 3

optional

As a verification, PowerFactory's own relay and fuse models reproduce the calculated operating times within 1.22 per cent, over 1896 operating times.

SLIDE 26

6. Conclusions and Future Work

about 1 min 15 s to read

6. Conclusions and Future Work	
- The DSDR method was implemented on the IEEE 13-node feeder in DigSILENT PowerFactory with manufacturer-based relay and fuse characteristics; the model meets the IEEE benchmark within about 2 %. - Without DG, 35 of 39 cells with the starting fuses, 39 of 39 after the fuse revision. - With the DG and a conventional R2, 24 DSDR alone, 25 fuse revision alone, 31 combined DSDR and fuse revision, 38 of 39 (zero margin). The remaining LG fault at 692 is limited by R2's maximum delayed dial. - With a three-cycle breaker time the 38 cells become 31, and 26 with a fuse safety margin as well. - The DSDR is necessary but not sufficient; it needs the fuse revision, and faults below R2 remain a limit. PowerFactory's relay and fuse models confirm the times within 1.22 %.	
Future work IEEE 34-node feeder: explicit directional element for R2; practical margins; setting-group change triggered by the DG state; inverter-based DG; hardware-in-the-loop tests.	

Para 1

To conclude. The DSDR method was implemented on the IEEE 13-node feeder in PowerFactory with manufacturer-based relay and fuse characteristics, and the model meets the IEEE benchmark within about 2 per cent.

Para 2

With the DG, a conventional R2 keeps only 24 of 39 cells. The combined DSDR and fuse revision raises this to 38 of 39 under the zero-margin criterion. The remaining line-to-ground fault at 692 is limited by the maximum delayed dial of R2.

Para 3

The DSDR is necessary but not sufficient. Alone it gives only 25 cells; it needs the fuse revision, and faults downstream of R2 remain a limit. With practical margins the result becomes 31 and 26 cells.

▷ Point to: Future work, at the bottom

Para 4

For future work: the IEEE 34-node feeder, an explicit directional element for R2, practical margins, a change of setting group triggered by the state of the DG, inverter-based DG, and hardware-in-the-loop tests.

SLIDE 27

7. References

about 15 s to read

7. References

- [1] M. Yousaf, A. Jallian, K. M. Mubang, D. Suterlo, "An adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in unbalanced distribution networks with distributed generation," IEEE Trans. Ind. Appl., vol. 58, no. 2, pp. 1831-1842, 2022.
- [2] W. H. Kersting, "Radial distribution test feeders," IEEE Trans. Power Syst., vol. 6, no. 3, pp. 975-985, 1991.
- [3] G. Burtropoul et al., "IEEE standard overcurrent characteristic equations for overcurrent relays," IEEE Trans. Power Del., vol. 14, no. 3, pp. 988-992, 1999.
- [4] A. Nassef et al., "A classification technique for recloser-fuse coordination in distribution systems with distributed generation," IEEE Trans. Power Del., vol. 27, no. 1, pp. 170-180, 2012.
- [5] M. Yousaf, K. M. Mubang, D. Suterlo, "Overcurrent protection scheme for the IEEE 13-node benchmark test feeder with improved selectivity," IEEE PES General Meeting, 2020.
- [6] W. H. Kersting, G. Shroek, "Short circuit analysis of IEEE test feeders," IEEE PES T&D, 2012.
- [7] DigSILENT GmbH, PowerFactory 2021 User Manual, 2021.
- [8] GE Multilin, 720731 Feeder Protection Relay - Instruction Manual (GEK-180291F), 2010.

27/27

Para 1

These are the main references. The method is from Yousaf and co-authors, published in 2022 in the IEEE Transactions on Industry Applications. The test feeder and its short-circuit benchmark are from Kersting.

SLIDE 32

Thank You

about 10 s to read

Thank You

Questions and Discussion

32/32

Para 1

That completes my presentation. Thank you for your attention. I am happy to take your questions.

Appendix slides — use only if you are asked

Slides 28 to 31 come after the references. Do not present them; open one when a question needs it.

SLIDE 28

Appendix: Comparison with the Reference Paper

about 55 s to read

Appendix: Comparison with the Reference Paper			
Aspect	This project	Reference paper [2]	Assessment
Rated branch currents	within 2 %	Table II	Very good
Fault levels	IEEE benchmark, 4.73 kA at head	5.41 kA, some branches higher	Lower
R1 worked example (4219 A)	0.006 / 1.503 s	0.007 / 1.502 s	Very good
Coordination with DG, conventional R2	24 of 39 cells	30 of 39 (Fig. 14)	Good (29 cells agree)
Coordination with the DSDR	38 of 39 cells	39 of 39 (Fig. 17)	Good: one cell at dial limit
Fuse current, LL at 646	3.67 kA	3.64 kA (Fig. 15)	Very good
R2 reverse current, LL at 646	1.11 kA	1.54 kA	Lower
EMT fuse operation	melts 1.25 s, clears 1.01 s	1.21 s (Fig. 10)	Close
DG model	Dynamic synchronous machine	Synchronous machine	Same

Para 1

This table compares my results with the reference paper. The rated branch currents agree within 2 per cent, and the worked example of R1 agrees almost exactly.

Para 2

My fault levels are lower: 4.73 kiloamperes at the feeder head against 5.41 in the paper, because I validated the model against the IEEE benchmark. With the DG and a conventional R2 the paper reports 30 coordinated cells and I obtained 24; 29 cells agree. With the DSDR the paper reports 39 and I obtained 38; the one cell is at the dial limit.

Para 3

So the main finding is reproduced: the DSDR with the fuse revision restores the coordination between the grid and the DG.

SLIDE 29

Appendix: Fault Levels - IEEE Benchmark, This Study and the Paper

about 50 s to read

Appendix: Fault Levels – IEEE Benchmark, This Study and the Paper				
Node	IEEE benchmark (kA)	This study (kA)	Study vs benchmark (%)	Paper vs benchmark (%)
632	4.80	4.74	-1.3 %	+17.7 %
632	4.80	4.80	-1.1 %	+62.3 %
645	3.41	3.42	+0.3 %	+24.6 %
646	3.05	3.11	+2.0 %	+19.3 %
671	3.35	3.39	+1.2 %	+34.9 %
692	3.35	3.39	+1.2 %	+10.1 %
695	3.12	3.10	-0.6 %	+100.0 %
696	3.90	3.90	+0.7 %	+114.8 %
694	2.62	2.66	+1.5 %	+32.4 %
692	1.79	1.87	+4.7 %	+49.2 %
611	1.85	1.86	+0.4 %	+4.3 %

Maximum fault current, DG not. Benchmark: Kesting & Dommel [6], values as quoted in [2] paper. Error of the branch voltage at the node.

- This study vs IEEE benchmark: within 2 % at 10 of 11 nodes, largest difference 4.7 % (652, LG).
- Paper vs benchmark: 4 % to 151 % higher.
- The paper's 5.73 kA (632-633) and 7.83 kA (692-676) exceeds its own possible in a radial feeder without DG.
- So the model was validated against the benchmark, not tuned to the paper's Table II.

Para 1

My fault levels are lower than the paper's, so I checked both against the IEEE short-circuit benchmark. My model agrees within 2 per cent at ten of the eleven nodes; the largest difference is 4.7 per cent at node 652.

Para 2

The paper's values are 4 to 151 per cent higher than the benchmark. Two of them, 6.73 and 7.83 kiloamperes, are higher than the paper's own value at the feeder head, 5.41 kiloamperes. That is not possible in a radial feeder without DG. So I validated the model against the benchmark and did not tune it to the paper's table.

